

HUNDRED DAYS OF CODE (DAY – 8)

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Date – 01/09/2026

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❖ Question 1:

Write a C program to find the volume and total surface area of a hemisphere in centimeters when its radius is given in meters.

We must create a C Program that will take radius in meter as input and calculate volume and total surface area of hemisphere in centimeter as output.

We will use the following formula to calculate volume and TSA(total surface area):

$$\text{Volume(V)} - \frac{2}{3} * \pi r^3$$

$$\text{Total surface area (TSA)} - 3\pi r^2$$

• Test Case Table:

Test case	Input	Expected Output	Actual Output	Result
1	R = 0.02	V = 16.75 TSA =37.68	V =16.75 TSA =37.68	True
2	R = 2	V = 16746667 TSA =376800	V = 16746667 TSA =376800	True
3	R = 0.068	V = 658.21 TSA = 435.58	V = 658.21 TSA =435.58	True

• Algorithm:

- 1) Start
- 2) Declare the variables r, v, TSA.
- 3) Read the radius of hemisphere in cm.
- 4) Calculate the Volume of hemisphere.

$$V = \frac{2}{3} * \pi * r^3$$

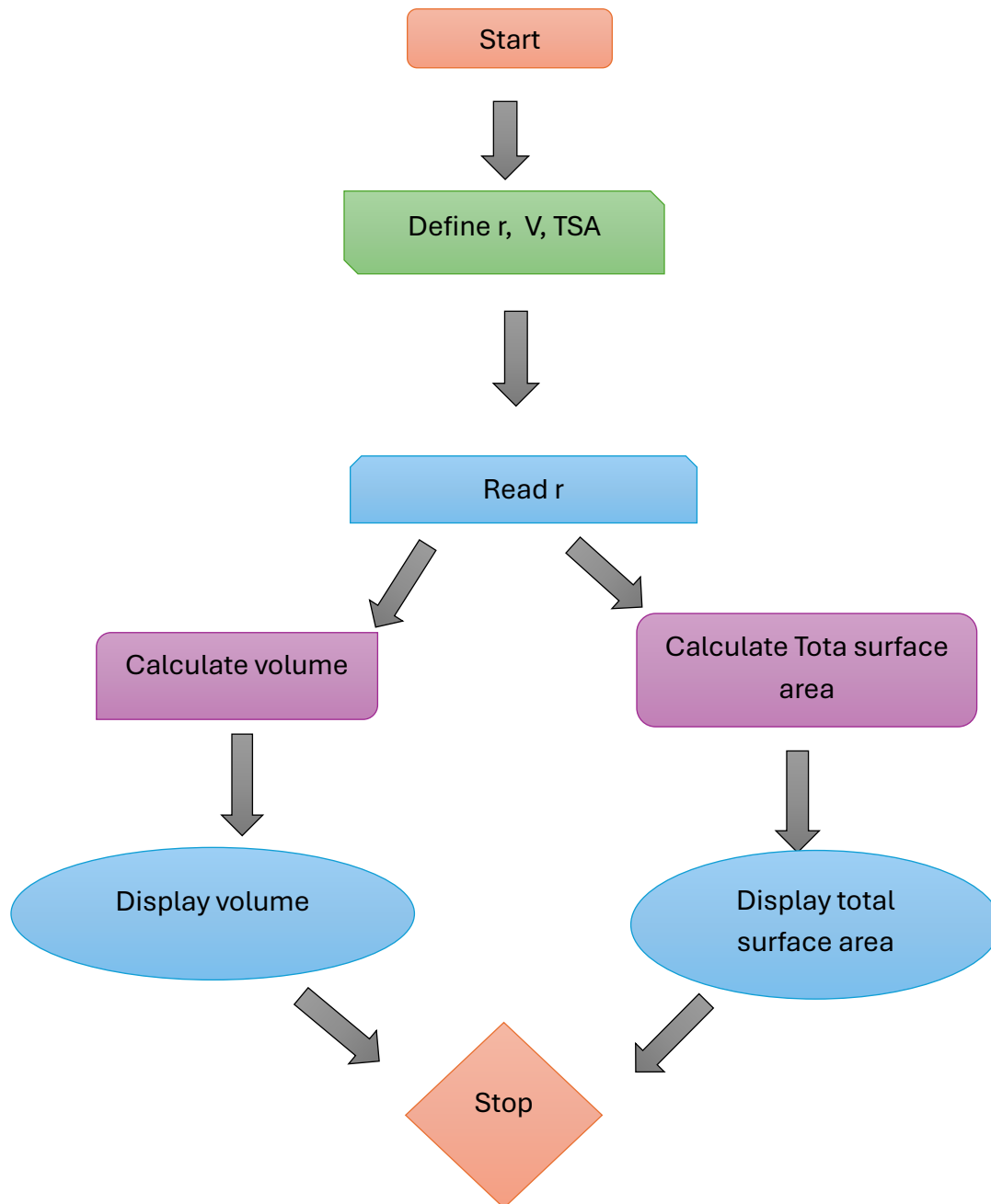
- 5) Calculate the TSA of hemisphere.

$$\text{TSA} = 3\pi * r^2$$

- 6) Display the value of volume in cm^3 .
- 7) Display the value of TSA in cm^2 .

8) stop.

- **Flowchart:**



- **C Program:**

```
#include<stdio.h>
#include<math.h>
#define PI 3.14
int main()
{
    float V, r, TSA;
    r = r*100;
    printf("Enter the radius of hemisphere in meters:");
    scanf("%f", &r);

    V = (2.0/3.0)*PI*r*r*r;

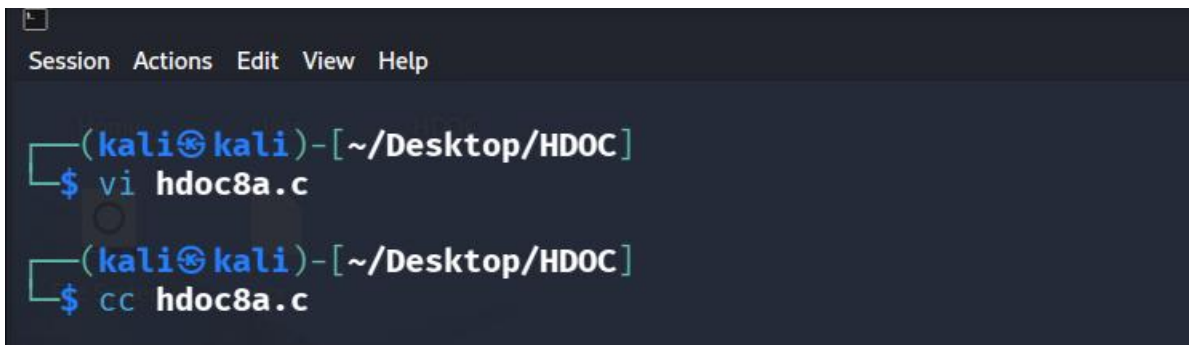
    TSA = 3*PI*r*r;

    printf("Volume of hemisphere = %.2f cubic cm\n", volume);
    printf("Total surface area of hemisphere = %.2f square cm\n", tsa);

    return 0;
}
```

Compile it with:

cc hdoc8a.c

A screenshot of a terminal window with a dark background. The window has a title bar with a small icon and the text "Session Actions Edit View Help". The terminal shows two commands being executed in a Kali Linux environment. The first command is "vi hdoc8a.c" and the second is "cc hdoc8a.c". The prompt is "(kali@kali) - [~/Desktop/HDOC]".

```
(kali@kali) - [~/Desktop/HDOC]
$ vi hdoc8a.c

(kali@kali) - [~/Desktop/HDOC]
$ cc hdoc8a.c
```

Run It using:

./a.out

```

(kali㉿kali)-[~/Desktop/HDOC]
$ ./a.out
Enter radius in meters: 0.02
Volume of hemisphere = 16.75 cubic cm
Total surface area of hemisphere = 37.68 square cm

(kali㉿kali)-[~/Desktop/HDOC]
$ ./a.out
Enter radius in meters: 2
Volume of hemisphere = 16746667.00 cubic cm
Total surface area of hemisphere = 376800.00 square cm

(kali㉿kali)-[~/Desktop/HDOC]
$ ./a.out
Enter radius in meters: 0.068
Volume of hemisphere = 658.21 cubic cm
Total surface area of hemisphere = 435.58 square cm

(kali㉿kali)-[~/Desktop/HDOC]
$ 

```

❖ QUESTION 2)

Write a C program to find the area and perimeter of an annulus when outer radius and inner radius are given.

First, we need to take the outer radius and inner radius of the annulus as input. Then, using these two radii, we calculate the area by subtracting the area of the inner circle from the area of the outer circle. Finally, we calculate the perimeter by adding the circumferences of the outer and inner circles.

We should use the following formulas:

$$\text{Area} = \pi(R^2 - r^2)$$

$$\text{Perimeter} = 2\pi(R + r)$$

Where, R = outer radius , r = inner radius

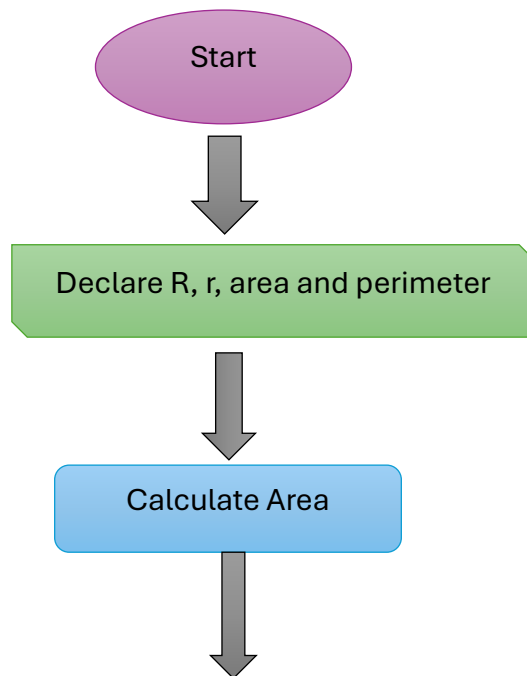
- **Test Case Table:**

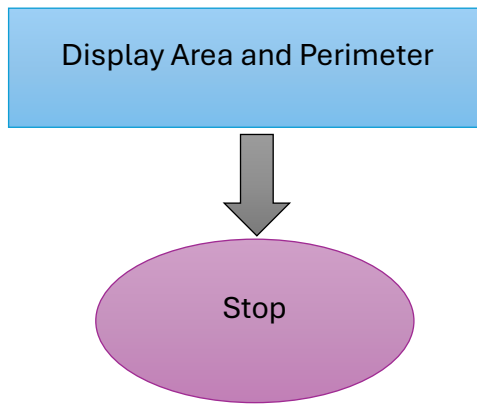
Test case	Input	Expected Output	Actual output	Result
1	R=12 r = 7	Area = 298.30 Perimeter = 119.32	Area = 298.30 Perimeter = 119.32	Pass
2	R= 6 r = 3	Area = 84.78 Perimeter = 56.52	Area = 84.78 Perimeter = 56.52	Pass
3	R= 4 r = 1	Area = 47.10 Perimeter = 31.40	Area = 47.10 Perimeter = 31.40	Pass

- **Algorithm:**

- 1) Start.
- 2) Declare variables R, r, area and perimeter.
- 3) Input the outer radius (R) and inner radius (r).
- 4) Calculate the area.
- 5) Calculate the perimeter.
- 6) Display the area and perimeter.
- 7) Stop.

- **Flowchart:**





- **C Program:**

```
#include <stdio.h>

#define pi 3.14

int main()
{
    float R, r ;
    printf("Enter outer radius: ");
    scanf("%f", &R);
    printf("Enter inner radius: ");
    scanf("%f", &r);
    float area = pi * (R * R - r * r);
    float perimeter = 2 * pi * (R + r);
    printf("Area of annulus = %.2f\n", area);
    printf("Perimeter of annulus = %.2f\n", perimeter);
    return 0;
}
```

Compile it with:

cc hdoc8b.c

```
Session Actions Edit View Help

(kali㉿kali)-[~/Desktop/HDOC]
$ vi hdoc8b.c

(kali㉿kali)-[~/Desktop/HDOC]
$ cc hdoc8b.c
```

Run it using:

`./a.out`

```
(kali㉿kali)-[~/Desktop/HDOC]
$ ./a.out
Enter outer radius: 12
Enter inner radius: 7
Area of annulus = 298.30
Perimeter of annulus = 119.32

(kali㉿kali)-[~/Desktop/HDOC]
$ ./a.out
Enter outer radius: 6
Enter inner radius: 3
Area of annulus = 84.78
Perimeter of annulus = 56.52

(kali㉿kali)-[~/Desktop/HDOC]
$ ./a.out
Enter outer radius: 4
Enter inner radius: 1
Area of annulus = 47.10
Perimeter of annulus = 31.40
```